ITACA: fast nonlinear CSA response

Generated from Julia validation results

2026-11-05

1 Model and domain

The state $x = V_0 - V_{\text{out}}$ follows

$$\dot{x} = g(x)I(t)/e - r(x), \qquad x(0) = 0.$$

The gain g (V per elementary charge) and recovery r (V/s) are continuous piecewise-linear tables fitted to 240 saved SPICE histories. Recovery vanishes at equilibrium. Each prediction follows the complete current history from baseline, including the quiet tail. RK4 steps end at current knots and acquisition-window boundaries, with a maximum step of 100 μs. The calibrated excursion extends to 0.675908 V and current to 77.798 fA, for the existing 0.5 mm square pad, 5 mm final gap, 3 m/s gap velocity and uniform mean arrivals within each frame. The existing 5 ms voltage average, three-frame inverse and stationary Gaussian temporal noise covariance are retained. The guard remains 0.63 V. Controls and out-of-domain histories use SPICE.

2 Independent validation

Calibration and validation use disjoint event IDs. Of 9391 held-out histories, 9390 lie inside the calibration domain and 1 requires SPICE. The worst corrected-bucket error is 0.00903 ENC against the 0.5 ENC criterion. The largest event-total difference is 3.829 ions. Halving the integration step changes the corrected estimates by at most 0.0000350 ENC. Median model runtime is 2.05 ms per history on this machine.

Frame (ms)	Histories	Fallbacks	Max (ENC)	Median (ENC)	Runtime (ms)
20	502	1	0.00454	0.00150	1.99
30	2898	0	0.00547	0.00089	2.17
40	5991	0	0.00903	0.00077	2.02

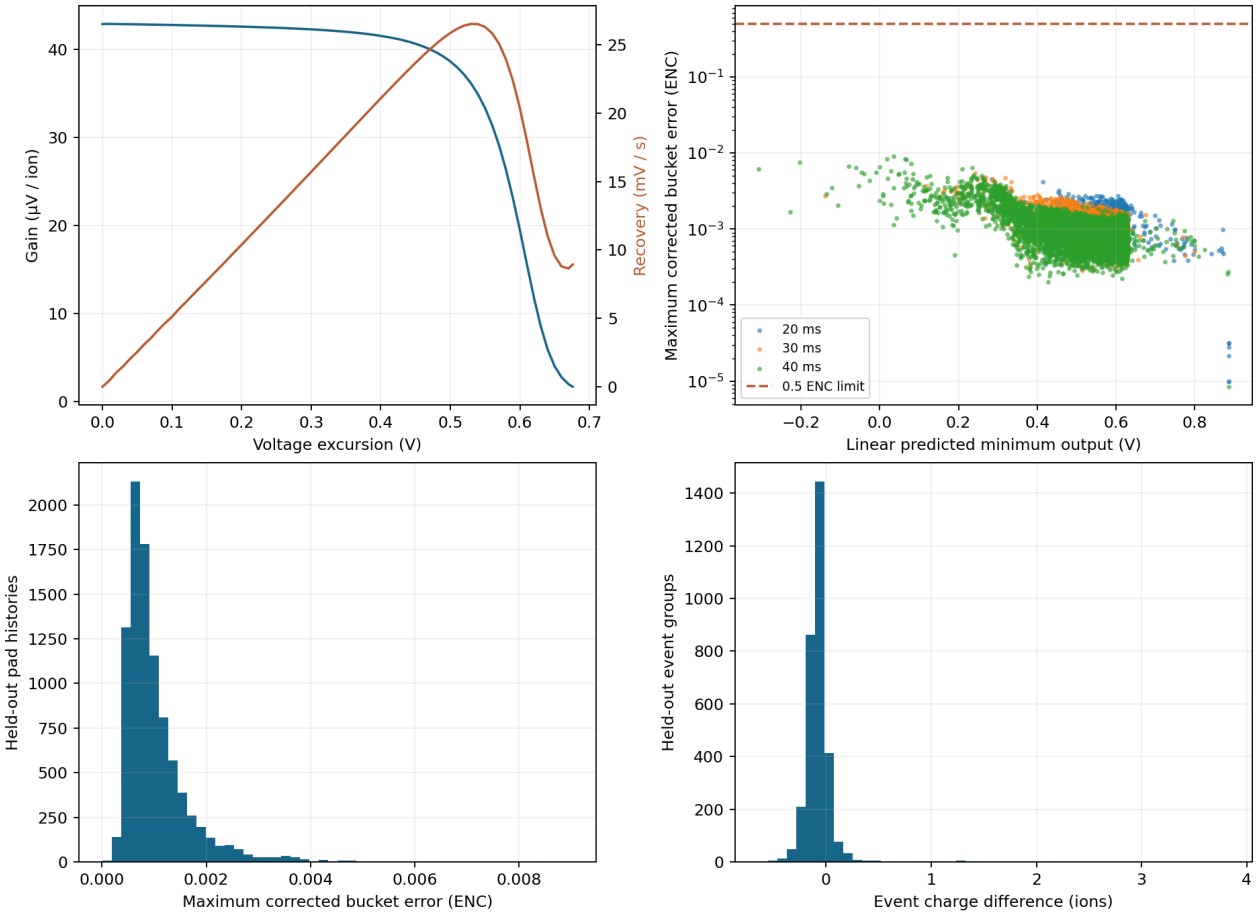


Figure 1: Fitted response and held-out validation. Event totals replace only below-guard histories inside the domain; unchanged linear and fallback SPICE histories cancel.

3 End-to-end production pilot

The 100-event 4 mm pilot has 8460 linear, 138 fast nonlinear and 469 SPICE-control histories. The nonlinear replacement alone differs from saved SPICE by at most 0.00655 ENC per bucket and 0.741 ions per event. The nonlinear evaluation stage, including control predictions, took 3.83 s. The filtered workflow took 104.0 s after preparation, including startup, SPICE controls, solver refinement and reduction. These timings do not include input preparation or the later correction and publication stages; they are not a full-production speedup measurement.

The smaller pilot selects different above-guard SPICE controls from the original 8,000-event run. Including that selection change gives a maximum bucket difference of 0.4154 ENC, maximum event-charge difference 199.018 ions and mean difference -53.175 ions. The maximum-pulse difference is 0.096 ions. The model-only metric isolates the nonlinear replacement. A common noise draw isolates response differences; new production identities otherwise have independent deterministic noise draws. Signed estimates, quiet buckets, covariance and ion ledgers pass the existing production checks.

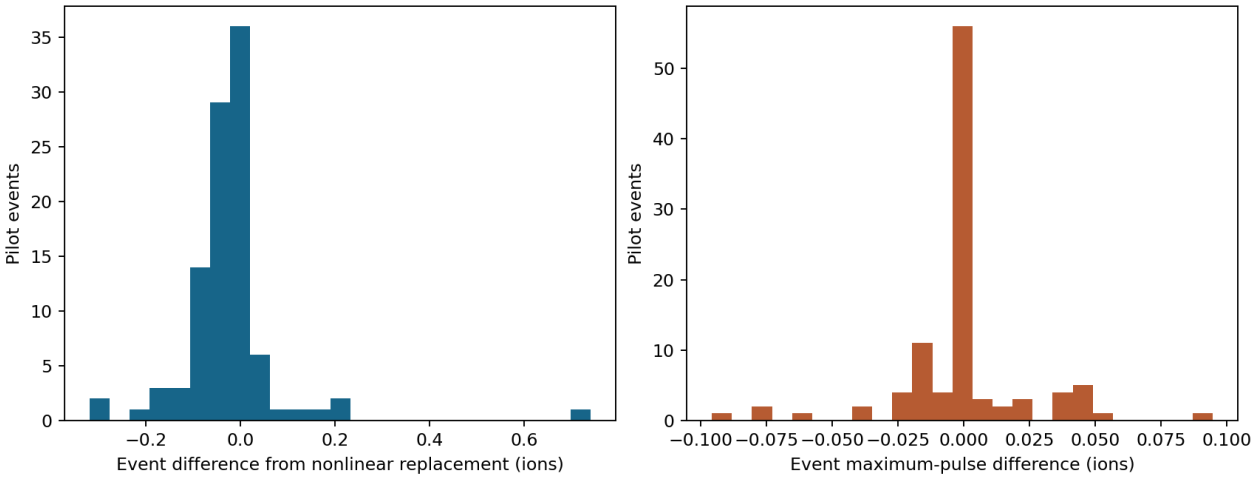


Figure 2: Model-only charge differences and complete-pipeline maximum-pulse differences.

4 Production and provenance

New production defaults to `hybrid_nonlinear`. An explicit `response_model = "hybrid_spice"` selects the reference backend. HDF5 model codes are 0 linear, 1 SPICE and 2 calibrated nonlinear. Calibration content, circuit and model code are checked and hashed into the resolved run identity. Numerical results and source hashes are in `results/ion_csa_nonlinear/v1/`. Reproduce the pilot with `make ion-csa-nonlinear-pilot`, and regenerate this report with `make ion-csa-nonlinear-report`. The numerical calibration is a local HDF5 artifact regenerated by the calibration script and its TOML card.